🎓 B.Tech – Computer Science & Engineering

Industrial Internship Report on

"QuickBite — Full Stack Food Delivery Platform"

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by Upskill Campus in collaboration with Industrial Partner UniConverge Technologies Pvt. Ltd. (UCT). This internship was focused on a Full Stack Development project provided by UCT. The project was completed including this report in 6 weeks' time.

My project was QuickBite — a Full Stack Food Delivery Web Application built using HTML5, CSS3, JavaScript, Node.js, and Express.js. The platform allows users to browse restaurants, search for food, manage a cart, place orders with unique Order IDs, track deliveries in real time, and manage their profiles. An Admin Dashboard was also implemented for managing orders and analytics.

This internship gave me excellent exposure to real-world full stack development practices including REST API design, JWT authentication, state management, and deployment using Vercel. It was an overall great learning experience.

Upskill Campus & UniConverge Technologies Pvt. Ltd.

Full Stack Development Internship | May – June 2026

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1. Preface

This report summarizes six weeks of intensive full stack web development work completed during my industrial internship at Upskill Campus in collaboration with UniConverge Technologies Pvt. Ltd. During this period, I designed, developed, and deployed QuickBite — a complete Food Delivery Web Application.

Internship opportunities are critical in today's competitive job market. They bridge the gap between academic learning and real-world industry practices. As a B.Tech CSE student, this internship gave me the practical experience needed to understand how large-scale web applications are architected, built, and deployed.

My project — QuickBite — is a full stack food delivery platform comparable in functionality to Swiggy and Zomato. It includes user authentication, restaurant browsing, food search, cart management, order placement with unique Order IDs, live order tracking, and an admin dashboard.

Upskill Campus provided the platform, learning materials, and weekly evaluation structure that guided me throughout the internship. UCT provided the real-world problem statement and industry context.

I am thankful to the Upskill Campus team, HR Manager Prabha Singh (UniConverge Technologies), and all mentors who supported me throughout. To my fellow students and juniors: embrace every internship opportunity — it will give you skills no textbook can fully prepare you for.

2. Introduction

2.1 About UniConverge Technologies Pvt. Ltd.

UniConverge Technologies Pvt. Ltd. (UCT) is a company established in 2013, working in the Digital Transformation domain and providing industrial solutions with a prime focus on sustainability and Return on Investment (RoI).

UCT leverages various cutting-edge technologies including Internet of Things (IoT), Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, and Frontend development.

Key Products & Platforms:

- UCT Insight — An IoT platform for quick deployment of IoT applications with analytics, alerts, dashboards, and third-party integrations (Power BI, SAP, ERP). Built in Java backend and ReactJS frontend.
- Smart Factory Platform (Factory Watch) — A platform for smart factory needs including OEE monitoring, predictive maintenance, and digital twin capabilities.
- LoRaWAN Solutions — Early adopter of LoRaWAN technology for AgriTech, Smart Cities, Smart Street Lights, and Smart Metering.
- Predictive Maintenance — Industrial machine health monitoring using Embedded Systems, Industrial IoT, and Machine Learning.

2.2 About Upskill Campus (USC)

Upskill Campus, along with The IoT Academy and in association with UniConverge Technologies, facilitates the smooth execution of the complete internship process. USC is a career development platform that delivers personalized executive coaching in an affordable, scalable, and measurable way.

Upskill Campus aims to upskill 1 million learners in the next 5 years by providing structured internship programs, industry expert interactions, and career growth services.

Website: https://www.upskillcampus.com/

2.3 Objectives of this Internship Program

The objectives of this internship program were:

- To gain practical experience working in an industry environment
- To solve real-world problems through software development
- To improve job prospects by building a production-quality project

- To develop a deeper understanding of full stack web development and its applications
- To achieve personal growth in communication, problem solving, and technical skills

2.4 References

[1] Node.js Official Documentation — https://nodejs.org/docs

[2] Express.js Official Documentation — https://expressjs.com/

[3] MDN Web Docs (HTML, CSS, JS) — https://developer.mozilla.org/

[4] JWT Authentication Guide — https://jwt.io/introduction

[5] GitHub Repository — https://github.com/VishnuRali/upskillcampus

2.5 Glossary

Term	Definition
HTML5	HyperText Markup Language version 5 — webpage structure
CSS3	Cascading Style Sheets version 3 — webpage styling
REST API	Representational State Transfer Application Programming Interface
JWT	JSON Web Token — used for secure user authentication
Node.js	JavaScript runtime built on Chrome V8 engine for server-side code
Express.js	Minimal web framework for Node.js
MongoDB	NoSQL document-oriented database
SPA	Single Page Application — loads one HTML page dynamically
MVC	Model-View-Controller software design pattern
Vercel	Cloud deployment platform for frontend applications

3. Problem Statement

The food delivery industry has grown significantly in India, with platforms like Swiggy and Zomato becoming household names. However, developing a production-quality food delivery application requires expertise across frontend development, backend APIs, database management, authentication, and real-time features.

The assigned problem statement for this internship was to build a Full Stack Food Delivery Application that addresses the following real-world requirements:

- Users need to register, login, and maintain their own session securely
- Users need to browse multiple restaurants and their menus without performance issues
- Users need to search for restaurants and food items instantly
- Users need a cart that persists across sessions and supports coupons and discounts
- Users need to place orders that generate unique trackable Order IDs
- Users need real-time order tracking showing delivery progress
- Admins need a dashboard to monitor orders, users, and revenue analytics

The challenge was to implement all of this functionality as a working full stack application within 6 weeks, using only free and open-source technologies.

4. Existing and Proposed Solution

4.1 Existing Solutions and Limitations

Existing food delivery platforms like Swiggy and Zomato are fully functional but:

- They are proprietary and closed-source — cannot be studied or modified
- They require large infrastructure (microservices, Kubernetes, CDN) not suitable for a student project
- They use React Native for mobile and complex React.js for web — high learning curve
- They require paid services (Google Maps API, payment gateways, cloud databases)

4.2 Proposed Solution — QuickBite

QuickBite is a full stack food delivery web application that replicates the core experience of Swiggy/Zomato using entirely free and open-source technologies. The solution:

- Uses Vanilla JavaScript (no heavy framework) for maximum control and learning
- Uses Node.js + Express.js for a lightweight, fast REST API backend
- Uses JSON file storage as database fallback — no paid database required
- Uses LocalStorage for client-side persistence — works offline too
- Simulates real-time order tracking using JavaScript timers
- Deployed on Vercel for free hosting

Code Submission (GitHub Link)

https://github.com/VishnuRali/upskillcampus

Report Submission (GitHub Link)

https://github.com/VishnuRali/upskillcampus/blob/main/QuickBiteApp_Vishnu_USC_UCT.pdf

5. Proposed Design / Model

QuickBite follows a classic Client-Server architecture with a Single Page Application (SPA) frontend communicating with a RESTful backend.

5.1 High Level Diagram

System Architecture Overview:

🖥️ **FRONTEND**	⚙️ **BACKEND**	🗄️ **DATABASE**
HTML + CSS + JS	Node.js + Express	MongoDB + JSON
Vercel	REST API + JWT	LocalStorage

5.2 Low Level Diagram — Backend File Structure

File / Folder	Purpose
server.js	Main entry point — sets up Express server, middleware, routes
config/db.js	MongoDB database connection configuration
controllers/authController.js	Register, login logic with JWT and bcrypt
controllers/ restaurantController.js	Fetch restaurants, menus, search
controllers/ orderController.js	Create orders, generate IDs, fetch history
controllers/userController.js	Profile, favorites, addresses
controllers/ adminController.js	Dashboard analytics, all orders and users
middleware/auth.js	JWT token verification middleware
models/User.js	MongoDB User schema
models/Order.js	MongoDB Order schema with auto-generated ID
models/Restaurant.js	MongoDB Restaurant and Menu schema
routes/*.js	5 route files mapping URLs to controllers
data/restaurants.json	12 restaurants with full menus (mock data)

5.3 Interfaces (Application Pages)

Login / Register Page: User authentication with demo accounts, email/password login

Home Page: Hero banner, food categories, trending restaurants, popular dishes, promo banners

Restaurants Page: All restaurants with search, sort, and cuisine filter

Restaurant Detail Page: Full menu with categories, Add to Cart buttons

Cart Page: Cart items, quantities, coupon input, address, order summary

Order Tracking Page: 6-stage live tracking timeline with progress animation

Order History Page: All previous orders with Track and Reorder buttons

Favorites Page: Saved restaurants with remove option

Addresses Page: Manage saved delivery addresses

Profile Page: User info, stats (orders, favorites, total spent)

Admin Dashboard: Stats, recent orders table, top selling items

6. Performance Test

Performance testing was conducted to verify that QuickBite meets real-world usability constraints.

6.1 Test Plan / Test Cases

#	Test Case	Expected Result	Outcome
1	Login with demo credentials	Redirect to home page with toast	✅ **PASS**
2	Search 'pizza'	Show Pizza Palace in suggestions	✅ **PASS**
3	Add item to cart	Cart badge increments, item appears	✅ **PASS**
4	Apply coupon FIRST50	₹50 discount applied to total	✅ **PASS**
5	Place order	Order ID QB-2026-XXXXXX generated	✅ **PASS**
6	Order tracking auto-progress	Stages update every 8-90 seconds	✅ **PASS**
7	Cart persists after refresh	Items still in cart on reload	✅ **PASS**
8	Admin dashboard loads	Shows orders, revenue, top items	✅ **PASS**
9	Mobile responsive layout	UI adapts to small screens	✅ **PASS**
10	Invalid login attempt	Error message shown, no redirect	✅ **PASS**

6.2 Test Procedure

- All tests were manually performed using a browser (Google Chrome)
- Backend tested using VS Code terminal running Node.js on localhost:5000
- Frontend tested using VS Code Live Server on localhost:5500
- Cart persistence tested by hard-refreshing the browser
- Order tracking tested by placing an order and waiting for status changes
- Mobile responsiveness tested using Chrome DevTools device emulator

6.3 Performance Outcome

All 10 test cases passed successfully. The application performs well with:

- Page load time under 1 second (no heavy frameworks)

- Search suggestions appearing within 300ms (debounced)
- Cart operations (add/remove/update) reflecting instantly
- Order ID generated uniquely every time (format: QB-2026-XXXXXX)
- All data persisting correctly across page refreshes using LocalStorage

7. My Learnings

This internship was transformational for my technical and professional growth. Here is a summary of my key learnings:

Technical Skills Gained:

- Full Stack Development — Building both frontend and backend of a production-quality web app
- REST API Design — Creating clean, organized API endpoints using Express.js
- JWT Authentication — Implementing secure login with JSON Web Tokens and bcryptjs
- State Management — Managing application state without any framework using pure JavaScript
- LocalStorage Persistence — Ensuring data survives browser refreshes on the client side
- MVC Architecture — Organizing code into Models, Views, and Controllers for scalability
- Git & GitHub — Version control, branching, and maintaining a professional repository
- Deployment — Deploying a real application on Vercel with a live URL
- Responsive Design — Making the UI work on mobile, tablet, and desktop
- Debugging — Identifying and fixing JavaScript, CSS, and API errors

Java & JDBC Learnings (Week 2 & 3):

- Java OOPs — Classes, Objects, Inheritance, Polymorphism, Abstraction, Encapsulation
- JDBC — Java Database Connectivity, Prepared Statements, 8-step connection process
- Java 8 Features — Lambda Expressions, Stream API, Optional Class, Date & Time API

Professional Skills Gained:

- Writing structured weekly progress reports
- Time management — completing 6 weeks of work on schedule
- Problem solving — overcoming technical challenges independently
- Documentation — writing clean README files and code comments

This internship helped me realize that building real-world applications requires much more than just coding. It requires planning, architecture, testing, documentation, and deployment — all of which I now have hands-on experience with.

8. Future Work Scope

Due to the 6-week time constraint, some advanced features could not be implemented. These are planned for future enhancement:

Phase 2 — Backend Enhancements:

- Full MongoDB integration with Mongoose ORM replacing JSON file storage
- Real-time order tracking using WebSockets (Socket.io)
- Payment gateway integration (Razorpay or Stripe)
- Email notifications for order confirmation and delivery updates
- Push notifications using Firebase Cloud Messaging

Phase 3 — Frontend Enhancements:

- Migrate to React.js for better component management and performance
- Implement Google Maps API for real delivery location tracking
- Add food item ratings and reviews system
- Implement advanced analytics charts using Chart.js or D3.js
- Progressive Web App (PWA) support for mobile installation

Phase 4 — Business Features:

- Multi-vendor system allowing restaurants to manage their own menus
- Delivery partner app for riders to accept and update orders
- Subscription plans for premium users (faster delivery, extra discounts)
- Machine learning based food recommendations using order history

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